EXERCISE

1. List the elements of the following sets.

(a) $A = \{x / x^3 - x = 0, x \in Z\}$.

(b) $B = \{x / x^3 - x = 0, x \in N\}$.

(c) $C = \{x / 2x + 3 = 4, x \in Z\}$.

(d) $D = \{x / x \in N, x$ is a multiple of 5 and $x^2 < 500\}$.

(e) $E = \{x / x$ is a two digit number whose sum of the digits is $7\}$.

Ans. : (a) $A = \{-1, 0, 1\}$ (b) $B = \{1\}$

(c) $C = \phi$. (d) $D = \{5, 10, 15, 20\}$

(e) $E = \{16, 25, 34, 43, 52, 61, 70\}$

2. Write the following sets by the property method

(a) $A = \{-2, 2\}$ (b) $B = \{1, 3, 5, 7, ...\}$

(c) $C = \{23, 29\}$ (d) $D = \{1, 2, 3, 4, 6, 12\}$

(e) $E = \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}\right\}$ (f) $F = \phi$.

Ans. : (a) $A = \{x / x^2 - 4 = 0, x \in Z\}$.

(b) $B = \{x / x = 2n - 1, n \in N\}$.

(c) $C = \{x / x$ is a prime number between 20 and 30$\}$.

(d) $D = \{x / x$ is a factor of 12$\}$.

(e) $E = \left\{\frac{n}{n+1} \Big/ n \leq 5, n \in N\right\}$.

(f) $F = \{x / x$ is a even prime number between 5 and 10$\}$.

3. Obtain the power set of $A = \{0, \phi\}$.

Ans. : $P(A) = \{\phi, \{0\}, \{\phi\}, A\}$.

4. Obtain the subsets of $\{x, y, z\}$.

Ans. : $\{x\}, \{y\}, \{z\}, \{x, y\}, \{y, z\}, \{x, z\}, \phi, \{x, y, z\}$.

5. Write the subsets of $\{\phi\}$.

Ans. : $\phi, \{\phi\}$.

6. $U = \{x / 1 \leq x \leq 10, x \in N\}$

$A = \{1, 2, 3, 4, 5, 6\}, B = \{2, 3, 5, 7\}$

Find $A', B', A \cup B, A \cap B, A - B, B - A$ and $A \Delta B$.

Ans. : $\{7, 8, 9, 10\}, \{1, 4, 6, 8, 9, 10\}, \{1, 2, 3, 4, 5, 6, 7\}, \{2, 3, 5\}, \{1, 4, 6\}, \{7\}, \{1, 4, 6, 7\}$ respectively.

7. $A = \{n / n \in N, n < 10\}$, $B = \{n / n \in N, n > 5\}$ and $C = \{3n / n \in N\}$ then find $A \cap B$ and $A \cap C$.

Ans. : $A \cap B = \{6, 7, 8, 9\}$, $A \cap C = \{3, 6, 9\}$.

8. $U = \{x / 1 \le x \le 10, x \in N\}$, $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 4, 5, 6\}$ then find $A \cup B$, $A \cup C$, $B \cup C$, $A \cap B$, $A \cap C$, $B \cap C$, $B - C$, $(A \cup B)'$, A', B', C', $A - B$, $A - C$, $C - A$ and $A \Delta C$.

Ans. : $A \cup B = \{1, 2, 3, 4, 6, 8\}$

$A \cup C = \{1, 2, 3, 4, 5, 6\}$

$B \cup C = \{2, 3, 4, 5, 6, 8\}$

$A \cap B = \{2, 4\}$, $A \cap C = \{3, 4\}$, $B \cap C = \{4, 6\}$

$B - C = \{2, 6\}$, $(A \cup B)' = \{5, 7, 9, 10\}$

$A' = \{5, 6, 7, 8, 9, 10\}$

$B' = \{1, 3, 5, 7, 9, 10\}$

$C' = \{1, 2, 7, 8, 9, 10\}$

$A - B = \{1, 3\}$, $A - C = \{1, 2\}$, $C - A = \{5, 6\}$,

$A \Delta C = \{1, 2, 5, 6\}$.

9. $U = \{1, 2, 3, ..., 10\}$, $A = \{1, 3, 5, 7, 9\}$, $B = \{1, 4, 5, 8\}$ and $C = \{2, 5, 6, 9\}$ then prove that

$$A \cap (B - C) = (A \cap B) - (A \cap C).$$

10. If $A = \{1, 2, 3\}$, $B = \{3, 4\}$ and $C = \{1, 4\}$ then show that

(a) $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

(b) $A \times (B - C) = (A \times B) - (A \times C)$.

(c) $A \times (B \Delta C) = (A \times B) \Delta (A \times C)$.

11. If $A = \{a, b, c\}$, $B = \{b, d\}$ and $C = \{c, d\}$, then prove that

(a) $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

(b) $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

(c) $A \times (B - C) = (A \times B) - (A \times C)$.

(d) $A \times (B \Delta C) = (A \times B) \Delta (A \times C)$.

12. In a class of 60 students, 24 students have failed in Maths, 18 have failed in English, 6 have failed in both. How many students passed in both the subjects ?

Ans. : 24.

13. In a class of 80 students, the number of students playing Cricket, Hockey and Football is denoted by the sets A, B and C. The following information is available : $n(A) = 30$, $n(B) = 40$, $n(C) = 50$, $n(A \cap B) = 15$, $n(B \cap C) = 17$, $n(C \cap A) = 23$ and $n(A \cap B \cap C) = 10$.

Find : (i) The no. of students playing exactly one game.

(ii) The no. of students playing exactly two games.

Ans. : (i) 40, (ii) 25.

14. In a class of 70 students, 20 students do not play Cricket or Hockey, 30 students play Cricket, 25 students play Hockey.

Find : (i) The no. of students playing both the games.

(ii) The no. of students playing only cricket.

(iii) The no. of students playing only hockey.

Ans. : (i) 5, (ii) 25, (iii) 20

15. 150 students sat in an examination of three subjects, Maths, Science and English. Out of them, 100 passed in Maths, 120 in Science and 80 in English. 85 passed in both Maths and Science, 70 in both Science and English and 60 in both English and Maths, and 50 students passed in all the three subjects.

Find : (i) The number of students failing in all the subjects

(ii) Number of students passing in exactly two subjects.

Ans. : (i) 5, (ii) 60.

16. If $n(U) = 100$, $n(A) = 32$, $n(B) = 42$, $n(A \cup B) = 62$, then find (i) $n(A' \cup B')$, (ii) $n(A \cup B')$

Ans. : (i) 80, (ii) 70.

17. $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $S = \{1, 3, 4\}$, $T = \{2, 4, 5\}$ then verify that

$$(A \times B) \cap (S \times T) = (A \cap S) \times (B \cap T).$$

18. In a school, 70 students play Cricket, 50 play Hockey, 40 play Volleyball. 19% students play Cricket and Hockey both, 13% students play both Cricket and Volleyball, whereas 15% play hockey and Volleyball both. 7% students play all the games. Prove that the above information is incorrect.

19. State whether the following statements are true of false ? Give reasons for your answer :

(1) $3 \in \{1, \{2, 3\}, 4\}$.

(2) $\phi = \{0\}$.

(3) Every set has atleast one proper subset.

(4) Every non - empty set has atleast two subsets.

(5) $A = \{x \in N / x^2 < 25\}$ has 8 subsets.

(6) $x \in A, A \in B$ then $x \in B$.

(7) $\{x, \{y, z\}\}$ has 9 subsets.

(8) The no. of subsets of $A \Delta A$ is 2.

(9) $\phi = \{\phi\}$.

(10) $\phi \in \phi$.

(11) If $A = B$ then $B \subset A$.

(12) $x \in A, A \subset B, B \subset C$ then $x \in C$.

(13) $a \in A, B \subset A, B \subset C$ then $a \in C$.

(14) $2 \subset \{1, 2, 3\}$.

(15) $A \cap \phi = A'$.

Ans. : False : 1, 2, 3, 5, 6, 7, 8, 9, 10, 13, 14, 15

True : 4, 11, 12.

20. For $A, B \in P(U)$, prove that

(i) $A \cup B = A \cap B \Leftrightarrow A = B$.

(ii) $A \subset B \Rightarrow B' \subset A'$.

(iii) $A = B \Rightarrow A' = B'$.

(iv) $(A \cap B) \cup (A \cap B') = A$.

(v) $(A \cup B) \cap (A \cup B') = A$.

(vi) $A \cap B = \phi \Rightarrow A \subset B'$.

(vii) $A \cap B = \phi, A \cup B = U$, then $B = A'$.

21. In a survey of 2000 students in a school, it was found that 48 % likes Mathematics, 54 % likes Physics and 64 % liked Chemistry. Of the total, 28 % liked Mathematics and Physics, 32 % liked Physics and Chemistry and 30% liked Mathematics and Chemistry. Only 6 % did not like any of these. Find

(i) The number of students liking all the three.

(ii) Physics and Chemistry but not Mathematics.

(iii) Only Mathematics.

Ans. : (i) 360, (ii) 280, (iii) 160.